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Multiple Choice - 2 points each unless otherwise indicated - circle your answer clearly. If you provide an explanation, the explanation may be used to assign partial credit if the answer you chose is incorrect.

- 1.) (2 pts) This question consists of several true/false questions (0.5 pts each) about the angular momentum of a quantum system. Circle whether each one is True or False.

- 2
- (T) / F For a system with a known angular momentum  $|\vec{L}| \neq 0$ , the maximum measurable value of  $L_z$  is less than  $|\vec{L}|$
- (T) / F It is possible to know both the  $z$  component of the angular momentum and the total angular momentum at the same time.
- T / (F) If  $|\vec{L}| \neq 0$ , it is possible to know both the  $x$  and  $y$  components of the angular momentum at the same time
- (T) / F For a hydrogen atom, it is possible to know both the atom's energy and its total angular momentum at the same time.

- 2.) We used the technique of separation of variables many times during this course. In one example we were separating the radial dependence of the wave function from the angular dependence with the guess  $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$ . We then derived

$$\frac{1}{R(r)} \frac{d}{dr} (r^2 R'(r)) - \frac{2mr^2}{\hbar^2} [V(r) - E] = \frac{-1}{Y} \left[ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] \quad (1)$$

We concluded that

- 2
- ✓ (a) Both sides of this equation must be zero
- (b) Both sides of this equation must be equal to the same constant which we called  $\ell(\ell + 1)$ .
- ✓ (c) The guess must be wrong because a function of  $r$  cannot be equal to a function of  $\theta$  and  $\phi$ .
- ✓ (d) The left side of this equation is equal to a constant which we called  $m$  and the right side of this equation is equal to a constant which we called  $\ell(\ell + 1)$
- ✓ (e) The left side of this equation is equal to zero and the right side of this equation is equal to a constant which we called  $\ell(\ell + 1)$

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3.) Using the canonical commutation relations for position and momentum, and the fact that

$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$ , determine the following commutator:

$$[\hat{L}_z, \hat{x}] = ? \quad (2)$$

(a)  $i\hbar\hat{y}$

(b)  $-i\hbar\hat{x}$

(c) 0

(d)  $-i\hbar\hat{p}_x$

(e)  $-i\hbar\hat{p}_y$

$$\begin{aligned} &= [\hat{x}\hat{p}_y, \hat{x}] - [\hat{y}\hat{p}_x, \hat{x}] \\ &= \hat{x}[\hat{p}_y, \hat{x}] + [\hat{x}, \hat{x}]\hat{p}_y - \left( \hat{y}[\hat{p}_x, \hat{x}] + [\hat{y}, \hat{x}]\hat{p}_x \right) \\ &= \hat{x}(-i\hbar) + 0 - \left( \hat{y}(i\hbar) + 0 \right) \\ &= -i\hbar\hat{y} - i\hbar\hat{y} \\ &= -2i\hbar\hat{y} \end{aligned}$$

4.) Consider two observables  $A$  and  $B$  and their associated operators  $\hat{A}$  and  $\hat{B}$ . In your lab, you perform the following experiment hundreds of times:

A particle is prepared in the state  $\psi_1$  (this is the same for all trials), you then make a single measurement of either  $A$  or  $B$ . The results of the experiment are as follows:

- Whenever  $A$  is measured, you always get the same value  $a_1$
- Whenever  $B$  is measured, sometimes you get value  $b_1$  and sometimes you get value  $b_2$ .

Are the following statements true or false? Circle your answer: (0.5 point each)

TRUE / FALSE

$\psi_1$  is an eigenfunction of  $\hat{A}$

TRUE / FALSE

$\psi_1$  is an eigenfunction of  $\hat{B}$

TRUE / FALSE

The operators  $\hat{A}$  and  $\hat{B}$  commute with each other

TRUE / FALSE

You cannot know both observables  $A$  and  $B$  at the same time.

Same eigen values  $\Leftrightarrow$  commute

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- 5.) For an electron in a hydrogen atom, how many different energy eigenfunctions have energy  $E_4 = -13.6\text{eV}/4^2 = -0.85\text{eV}$ ? In other words, what is the degeneracy of the  $n = 4$  energy level for hydrogen?

$$0 \leq l \leq n-1$$

$$-l \leq m \leq l$$

For  $n$ : have  $n$  l's,  $2l+1$  m's per l

$$\begin{aligned} \sum_{l=0}^{n-1} 2l+1 &= \sum_{l=1}^n 2l-1 = -n + 2 \frac{n(n+1)}{2} \\ &= -n + n^2 + n \\ &= n^2 \end{aligned}$$

- 6.) (4 pts) For each statement, mark whether it correctly applies to the one-dimensional infinite square well (ISW), the one-dimensional quantum harmonic oscillator (QHO), the hydrogen atom, or none of the above? Place a checkmark in the appropriate boxes. Note that it is possible for a statement to apply to more than one system.

| Statement                                      | ISW | QHO | Hydrogen | None of these |
|------------------------------------------------|-----|-----|----------|---------------|
| Ground state is labeled by $n = 1$             | ✓   |     | ✓        |               |
| Ground state energy is $E = 0$                 |     | ✓   |          |               |
| Energy levels are equally spaced               |     | ✓   |          |               |
| Energy eigenfunctions use Laguerre polynomials |     |     | ✓        |               |

Free Response Questions - Show all steps of your solution to receive credit. You may use the formulas and integrals on the attached formula sheet without proving them.

- 1.) (10 pts) A quantum particle of mass  $M$  experiences a harmonic oscillator potential  $V(x) = \frac{1}{2}M\omega^2 x^2$ . At time  $t = 0$ , the particle is in an equal superposition of ground state and first excited state. In other words, at time  $t = 0$  if you measured the energy of the particle, you would have a  $(2/3)$  probability of getting the ground state energy and a  $(1/3)$  probability of getting the first excited state energy.

4 (a) (4 pts) Write out the normalized initial wave function  $\Psi(\xi, 0)$  using the dimensionless variable  $\xi$  instead of  $x$ . Your expression should be in terms of  $\xi$ ,  $M$ ,  $\omega$ , and fundamental constants. If you use any other symbols, make sure they are defined clearly. It is ok to assume  $\Psi(\xi, 0)$  is real and positive for all  $\xi$ .

4 (b) (4 pts) What is the wave function at a later time  $\Psi(\xi, t)$ ? Your expression in terms of  $\xi$ ,  $M$ ,  $\omega$ ,  $t$  and fundamental constants. If you use any other symbols, make sure they are defined clearly.

2 (c) (2 pts) Does the probability density for this particle depend on time, or not? You can either explain your answer in words, or do a short calculation to justify your answer.

✓ a.)  $\Psi(\xi, 0) = \sqrt{\frac{2}{3}} \psi_0 + \sqrt{\frac{1}{3}} \psi_1 = \sqrt{\frac{2}{3}} \frac{\alpha}{\sqrt{2\pi}} \cdot 1 \cdot e^{-\xi^2/2} + \sqrt{\frac{1}{3}} \frac{\alpha}{\sqrt{2\pi}} \cdot 2\xi \cdot e^{-\xi^2/2}$

$$\boxed{\Psi(\xi, 0) = \left(\frac{M\omega}{\pi\hbar}\right)^{1/4} \left[ \sqrt{\frac{2}{3}} e^{-\xi^2/2} + \sqrt{\frac{1}{3}} \sqrt{2} \xi e^{-\xi^2/2} \right]}$$

✓ b.)  $\Psi(\xi, t) = \sum c_n \psi_n(\xi) e^{-iE_n t/\hbar} = \left(\frac{M\omega}{\pi\hbar}\right)^{1/4} \left[ \sqrt{\frac{2}{3}} \exp\left(-\frac{\xi^2}{2} - \frac{i\omega t}{2}\right) + \sqrt{\frac{1}{3}} \sqrt{2} \xi \exp\left(-\frac{\xi^2}{2} - \frac{3i\omega t}{2}\right) \right]$

$E_0/\hbar = \frac{\omega}{2}, E_1/\hbar = \frac{3\omega}{2}$

$$= \Psi(\xi, t)$$

✓ c.) Yes,  $\Psi(\xi, t) = A e^{iCt} + B e^{iDt}$

$$|\Psi|^2 = \Psi \Psi^* = (A e^{iCt} + B e^{iDt}) (A e^{-iCt} + B e^{-iDt}) = A^2 + B^2 + 2AB \left( \frac{e^{i(C-D)t} + e^{i(D-C)t}}{2} \right)$$

$$= A^2 + B^2 + 2AB \cos((C-D)t) = A(\xi)^2 + B(\xi)^2 + \underline{\underline{2A(\xi)B(\xi) \cos((C-D)t)}}$$

$C, D$  constant

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- 2.) (7 pts) An electron is in an infinite square well of length  $L$ . Initially, the electron's normalized wave function is:

$$\Psi(x, 0) = \begin{cases} \sqrt{\frac{30}{L}} \left(\frac{x}{L}\right) \left(1 - \frac{x}{L}\right) & 0 \leq x \leq L \\ 0 & x < 0 \text{ or } x > L \end{cases} \quad (3)$$

- (a) (2 pts) What is  $\hat{H}\Psi(x, 0)$ ? [Recall  $\hat{H} = \hat{p}^2/2M + V(\hat{x})$ .]  
 (b) (1 pt) Is  $\Psi(x, 0)$  an eigenfunction of  $\hat{H}$ ? Explain why or why not.  
 (c) (4 pts) Determine  $\langle p \rangle$  for this wave function (at  $t = 0$ ). Justify your answer with an explicit calculation.

✓ a.)  $-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = -\frac{\hbar^2}{2M} \sqrt{\frac{30}{L}} \frac{\partial^2}{\partial x^2} \left(\frac{x}{L} - \frac{x^2}{L^2}\right) = -\frac{\hbar^2}{2M} \cdot \frac{2}{L^2} = -\frac{\hbar^2}{ML^2}$

$$\hat{H}\Psi(x, 0) = \begin{cases} \frac{\sqrt{30}\hbar^2}{ML^{3/2}} & 0 \leq x \leq L \\ 0 & \text{else} \end{cases}$$

✓ b.) No:  $\Psi(x, 0)$  is not of the form  $\begin{cases} \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) & 0 \leq x \leq L \\ 0 & \text{else} \end{cases}$

So,  $\hat{H}\Psi \neq E_n\Psi$  for all values of  $n$

✓ c.)  $\langle p \rangle = \int_{\mathbb{R}} \Psi(x, 0)^* \hat{p} \Psi(x, 0) dx = \int_{\mathbb{R}} \sqrt{\frac{30}{L}} \left(\frac{x}{L}\right) \left(1 - \frac{x}{L}\right) (-i\hbar) \sqrt{\frac{30}{L}} \left(\frac{1}{L} - \frac{2x}{L^2}\right) dx$   
 $= -\frac{30i\hbar}{L^2} \int_0^L \left(\frac{x}{L}\right) \left(1 - \frac{x}{L}\right) \left(1 - \frac{2x}{L}\right) dx$   
 $= -\frac{30i\hbar}{L} \int_0^1 u(1-u)(1-2u) du$   
 $= -\frac{30i\hbar}{L} \int_0^1 (2u^3 - 3u^2 + u) du$   
 $= -\frac{30i\hbar}{L} \left(\frac{2}{4} - \frac{3}{3} + \frac{1}{2}\right) = 0$   
 $\Rightarrow \langle p \rangle = 0$

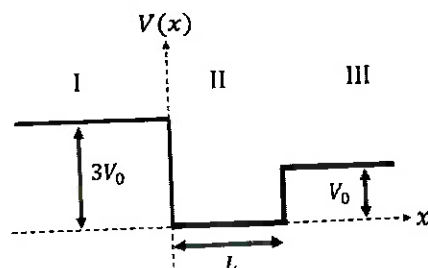
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- 3.) (9 pts) A quantum particle of mass  $M$  moves in the potential shown. The potential  $V(x < 0) = 3V_0$  and  $V(x > L) = V_0$ . The particle is in the lowest energy state with  $E < V_0$ , this is a bound state.



The wave function in region II is

$$\psi_{II}(x) = C \cos(kx) + D \sin(kx) \quad (4)$$

where  $k = \sqrt{2ME/\hbar^2}$

- 3 (a) (3 pts) What is the general form of the wave function in region I that solves the time-independent Schrödinger equation? Impose the requirement that the wave function must be normalizable (but you do NOT have to normalize it). Your expression should be in terms of  $E, M, \hbar, V_0, x$ , and one arbitrary constant, which we'll call  $A$ .
- 3 (b) (3 pts) What is the general form of the wave function in region III that solves the time-independent Schrödinger equation? Impose the requirement that the wave function must be normalizable (but you do NOT have to normalize it). Your expression should be in terms of  $E, M, \hbar, V_0, x$ , and one arbitrary constant, which we'll call  $F$ .
- 3 (c) (3 pts) By imposing appropriate boundary conditions at  $x = 0$ , determine  $C$  and  $D$  in terms of  $A, V_0$ , and  $E$ .

a.) ~~Assume particle is incident from left:~~  $-\frac{\hbar^2}{2m} \psi_I'' + 3V_0 \psi_I = E \psi_I \Rightarrow \psi_I = \frac{2m}{\hbar^2} (3V_0 - E) \psi_I$   
 Let  $q^2 = \frac{2m}{\hbar^2} (3V_0 - E)$ . Then,  $\psi_I = A e^{qx} + B e^{-qx}$ ,  $\psi(-\infty) = 0 \Rightarrow \boxed{\psi_I = A e^{qx}}$  ✓

b.)  $\frac{\hbar^2}{2m} \psi_{III}'' + V_0 \psi_{III} = E \psi_{III} \Rightarrow \psi_{III}'' = \frac{2m}{\hbar^2} (V_0 - E) \psi_{III}$   
 Let  $\gamma^2 = \frac{2m}{\hbar^2} (V_0 - E)$ . Then,  $\psi_{III} = F e^{-\gamma x} + G e^{\gamma x}$ ,  $\psi(\infty) = 0 \Rightarrow \boxed{\psi_{III} = F e^{-\gamma x}}$  ✓

c.)  $\psi_I(0) = \psi_{II}(0) \Rightarrow \boxed{A = C}$   
 $\psi_I'(0) = \psi_{II}'(0) \Rightarrow qA = kD \Rightarrow D = \frac{q}{k} A = \frac{\sqrt{\frac{2m}{\hbar^2} (3V_0 - E)}}{\sqrt{\frac{2m}{\hbar^2} E}} A = \boxed{\sqrt{\frac{3V_0 - E}{E}} A = D}$  ✓

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- 4.) (9 pts) A hydrogen atom is described by the wave function at time  $t = 0$

$$\Psi(\vec{r}, 0) = \frac{1}{\sqrt{96\pi a^3}} \left(\frac{r}{a}\right) e^{-r/2a} \quad (5)$$

which is NOT an energy eigenfunction. Here  $a$  denotes the Bohr radius.

- (a) (4 pts) Show that the state  $\Psi(\vec{r}, 0)$  is normalized.

- (b) (5 pts) If, at time  $t = 0$ , someone measures the energy of this atom, what is the probability that they will receive a value of  $-13.6\text{eV}$ ?

$$u = \frac{r}{a}$$

$$du = \frac{1}{a} dr$$

4 a.)  $\int_{\mathbb{R}^3} |\Psi(\vec{r}, 0)|^2 dV = 4\pi \int_0^\infty \frac{1}{96\pi a^3} \frac{r^2}{a^2} e^{-r/a} r^2 dr = \frac{1}{24a^5} \int_0^\infty r^4 e^{-r/a} dr$

$$= \frac{1}{24a^5} a^4 \int_0^\infty u^4 e^{-u} du = \frac{1}{24} 4! = \underline{\underline{1}}$$

5 b.)  $c_1 = \int_{\mathbb{R}^3} \Psi(\vec{r}, 0) \psi_{100}^*(r, \theta, \phi) dV = \int_{\mathbb{R}^3} \frac{1}{\sqrt{46\pi a^3}} \left(\frac{r}{a}\right) e^{-r/2a} \frac{1}{\sqrt{\pi a^3}} e^{-r/a} dV$

$$= \frac{1}{\pi a^3} \frac{1}{\sqrt{46}} \frac{1}{a} 4\pi \int_0^\infty r^3 e^{-\frac{3r}{2a}} dr \quad u = \frac{3r}{2a} \quad du = \frac{3}{2a} dr$$

$$= \frac{4}{\sqrt{46}} \frac{1}{a^4} \left(\frac{2a}{3}\right)^3 \left(\frac{2a}{3}\right) \int_0^\infty u^3 e^{-u} du = \frac{1}{\sqrt{46}} \frac{1}{24} \frac{16a^4}{81} \cdot 3! = \frac{1}{\sqrt{6}} \frac{16}{81} \cdot 6 = \frac{16\sqrt{6}}{81}$$

$$\hookrightarrow \sqrt{12.8} = \sqrt{6 \cdot 16} = 4\sqrt{6}$$

$$c_1 = \frac{16\sqrt{6}}{81} \quad P(E = E_1 = -13.6\text{eV}) = |c_1|^2 = \boxed{\frac{512}{2187}}$$

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5.) (6 pts) Consider the following wave function describing an electron in a hydrogen atom

$$\Psi = \frac{1}{2}\psi_{3,1,0} + \frac{i}{\sqrt{2}}\psi_{2,1,1} - \frac{1}{2}\psi_{3,2,0} \quad (6)$$

where we're using the usual notation  $\psi_{n,\ell,m}$  to denote the energy eigenfunctions for hydrogen.

3 (a) (3 pts) What is the expectation value of the z-component of the angular momentum  $\langle L_z \rangle$  for a particle described by this wave function?

3 (b) (3 pts) What is the expectation value of the total angular momentum squared  $\langle \vec{L}^2 \rangle$  for a particle described by this wave function?

$$\begin{aligned} a.) \langle L_z \rangle &= \int_{\mathbb{R}^3} \Psi^* \hat{L}_z \Psi dV = \int_{\mathbb{R}^3} \left( \frac{1}{2} \psi_{3,1,0}^* - \frac{i}{\sqrt{2}} \psi_{2,1,1}^* - \frac{1}{2} \psi_{3,2,0}^* \right) \left( 0 + \frac{i}{\sqrt{2}} \cdot 1\hbar \psi_{2,1,1} - \frac{1}{2} \cdot 0 \right) dV \\ &= \int_{\mathbb{R}^3} \frac{1}{2} \hbar |\psi_{2,1,1}|^2 + \text{cross terms} dV \end{aligned}$$

$$(\hat{L}_z \psi_{\ell}^m = m\hbar \psi_{\ell}^m \Rightarrow \hat{L}_z \psi_{n\ell m} = m\hbar \psi_{n\ell m})$$

$$(\text{similarly, } \hat{L}^2 \psi_{n\ell m} = \hbar^2 \ell(\ell+1) \psi_{n\ell m})$$

$$\boxed{\langle L_z \rangle = \frac{1}{2} \hbar}$$

$$\begin{aligned} b.) \langle L^2 \rangle &= \int_{\mathbb{R}^3} \Psi^* \hat{L}^2 \Psi dV = \int_{\mathbb{R}^3} \left( \frac{1}{2} \psi_{3,1,0}^* - \frac{i}{\sqrt{2}} \psi_{2,1,1}^* - \frac{1}{2} \psi_{3,2,0}^* \right) \hbar^2 \left( \frac{1}{2} (1)(2) \psi_{3,1,0} + \frac{i}{\sqrt{2}} (1)(2) \psi_{2,1,1} \right. \\ &\quad \left. - \frac{1}{2} (2)(3) \psi_{3,2,0} \right) dV \end{aligned}$$

$$= \hbar^2 \int_{\mathbb{R}^3} \frac{1}{2} |\psi_{3,1,0}|^2 + |\psi_{2,1,1}|^2 + \frac{3}{2} |\psi_{3,2,0}|^2 + \text{cross terms} dV$$

$$= \hbar^2 \left( \frac{1}{2} + 1 + \frac{3}{2} \right) \Rightarrow \boxed{\langle L^2 \rangle = 3\hbar^2}$$

THE END